

Scheme 6

BPS Ontology and Semantic Loading Benchmark Scheme

Ontology prompts for benchmark-relative semantic quantification

Purpose

This scheme supplies the ontology scaffold used to quantify semantic emphasis across biological, psychological, and social axes. It is not a manual adjudication form, but it is an operational text-classification framework because it standardizes the domain and subdomain prompts against which review records are embedded and compared.

Operational Status

- Workflow position: semantic loading and synthesis stage after Stage 2 coding.
- Operational mode: ontology-prompted embedding benchmark, with OpenRouter embeddings when available and TF-IDF fallback otherwise.
- Canonical implementation basis: `src/bps_review/reporting/semantic_loading.py` and the archived ontology terms JSON.

Canonical Source Paths

- `src/bps_review/reporting/semantic_loading.py`
- `src/vector_db/semantic_loading/ontology/ontology_terms.json`
- `src/vector_db/semantic_loading/analysis/domain_loading_summary.csv`
- `src/vector_db/semantic_loading/analysis/subdomain_loading_summary.csv`

Record Text Used for Scoring

Each review record is composed into a semantic analysis string with the following sections: title, abstract, and extracted objective text. The ontology scheme then compares that composed record text against the benchmark prompts below.

Prompt Templates

`domainprompt` `{domain} chronic pain ontology. {joined subdomains}.`
`subdomainprompt` `{domain} chronic pain subdomain. {term}.`

Domain Order

The benchmark axes are always ordered as `biological`, `psychological`, and `social`.

Ontology Content

Biological Subdomains

- Central Sensitization and Neuroplasticity
- Musculoskeletal and Structural Pathology
- Pharmacological and Biomedical Treatment
- Sleep Disruption and Circadian Dysregulation
- Immune Inflammatory and Neuroinflammatory Processes
- Genetic Epigenetic and Biological Vulnerability
- Neuroimaging Brain Structure and Function
- Nociceptive Transmission and Pain Pathways
- Physical Function Mobility and Deconditioning
- Metabolic Nutritional and Hormonal Factors

Psychological Subdomains

- Catastrophizing and Negative Cognitive Appraisal
- Fear Avoidance and Pain Related Fear
- Depression Emotional Distress and Affect
- Anxiety and Psychological Reactivity
- Self Efficacy Control Beliefs and Perceived Mastery
- Acceptance Psychological Flexibility and Mindfulness
- Pain Coping Strategies and Adjustment
- Attention Vigilance and Pain Processing
- Illness Beliefs Pain Representations and Meaning
- Cognitive Behavioral and Psychotherapeutic Approaches
- Third Wave Therapies ACT and Contextual Approaches
- Resilience Positive Psychology and Post Traumatic Growth
- Identity Self Concept and Chronic Pain Biography
- Trauma Adverse Childhood and Life Events
- Personality Psychological Traits and Individual Differences
- Cognitive Function Executive Processes and Brain Health

- Motivational Processes Goal Pursuit and Engagement
- Healthcare Seeking Treatment Adherence and Engagement
- Emotional Regulation and Pain Affect Processing
- Mental Health Comorbidity and Psychological Wellbeing

Social Subdomains

- Social Support Network and Interpersonal Resources
- Work Disability Occupational Function and Productivity
- Family Caregiver and Household Dynamics
- Socioeconomic Status and Health Inequity
- Healthcare Access Navigation and System Factors
- Cultural Ethnic and Demographic Context
- Community Participation and Social Role Functioning
- Legal Compensation and Medicolegal Systems
- Health Literacy Education and Patient Empowerment
- Stigma Social Isolation and Exclusion
- Return to Work Vocational Rehabilitation and Employment
- Social Determinants of Pain and Environment

Outputs Produced from This Scheme

- Record-level domain loadings.
- Record-level subdomain loadings.
- Domain summary and subdomain summary tables.
- Pairwise loading analyses.
- Dominance-by-review-type summaries.

Interpretive Note

This dossier is included because the project uses ontology prompts as a standardized semantic coding scaffold. It should be understood as an analytic coding scheme rather than a human extraction codebook.